



Multiple Choice:

- Given $h(x) = \sqrt{x+7}$ and $j(x) = 7x^2 - 3$. Express $(j \circ h)(x)$ in terms of x .
A. $6x + 32$ C. $7x + 46$
B. $8x + 48$ D. $5x + 53$
- In a class of 40 students, 27 students like Calculus and 25 like Geometry. How many students like both Calculus and Geometry?
A. 9 C. 11
B. 10 D. 12
- Let S be the collection of all sets with at most 5 elements. Then:
A. An element of S is a set with 1, 2, 3, 4, or 5 elements.
B. An element of S is a number.
C. An element of S is a set of 25 elements.
D. An element of S is a superset of S .
- Let $A = \{x, y\}$ and $B = \{x, y, z\}$. What is $A \cup B$?
A. B C. $\{x\}$
B. A D. $\{x, z\}$
- Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$. What is $B \times A$?
A. $\{(1,a), (1,b), (2,a), (2,b), (3,a), (3,b)\}$
B. $\{(a,1), (a,2), (a,3), (b,1), (b,2), (b,3)\}$
C. $\{(a,1), (2,a), (a,3), (b,1), (2,b), (b,3)\}$
D. $\{(1,a), (1,b), (a,2), (b,2), (3,a), (b,3)\}$
- A _____ is an ordered collection of objects.
A. relation C. function
B. set D. proposition
- The set O of odd positive integers less than 10 can be expressed by _____.
A. $\{1, 5, 7, 9, 11\}$ C. $\{1, 2, 5, 9\}$
B. $\{1, 3, 5, 7, 9\}$ D. $\{1, 2, 3\}$
- What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b\}$?
A. $\{(1, a), (1, b), (2, a), (b, b)\}$
B. $\{(1, 1), (2, 2), (a, a), (b, b)\}$
C. $\{(1, a), (2, a), (1, b), (2, b)\}$
D. $\{(1, 1), (a, a), (2, a), (1, b)\}$
- Which of the following two sets are equal?
A. $A = \{1, 2\}$ and $B = \{1\}$
B. $A = \{1, 2\}$ and $B = \{1, 2, 3\}$
C. $A = \{1, 2, 3\}$ and $B = \{2, 1, 3\}$
D. $A = \{1, 2, 4\}$ and $B = \{1, 2, 3\}$
- The set of positive integers is _____.
A. infinite C. finite
B. subset D. empty
- The members of the set $S = \{x \mid x \text{ is the square of an integer and } x < 100\}$ is:
A. $\{0, 2, 4, 5, 9, 58, 49, 56, 99, 12\}$
B. $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$
C. $\{1, 4, 9, 16, 25, 36, 64, 81, 85, 99\}$
D. $\{0, 1, 4, 9, 16, 25, 36, 49, 64, 121\}$
- The union of the sets $\{1, 2, 5\}$ and $\{1, 2, 6\}$ is the set _____.
A. $\{1, 2, 6, 1\}$ C. $\{1, 2, 5, 6\}$
B. $\{1, 2, 1, 2\}$ D. $\{1, 5, 6, 3\}$
- The intersection of the sets $\{1, 2, 5\}$ and $\{1, 2, 6\}$ is the set _____.
A. $\{1, 2\}$ C. $\{5, 6\}$
B. $\{2, 5\}$ D. $\{1, 6\}$
- Two sets are called disjoint if there _____ is the empty set.
A. union C. difference
B. intersection D. complement
- The difference of $\{1, 2, 3\}$ and $\{1, 2, 5\}$ is the set _____.
A. $\{1\}$ C. $\{5\}$
B. $\{3\}$ D. $\{2\}$
- Let A, B and C represent people who like apples, bananas, and carrots respectively. The number of people in $A = 10, B = 12$ and $C = 16$. Three people are such that they enjoy apples, bananas as well as carrots. Two of them like apples and bananas. Let three people like apples and carrots. Also, four people are such that they like bananas and carrots. Answer the following questions: How many people like apples only?
A. 2 C. 7
B. 4 D. 11
- If $n(A)=20$ and $n(B)=30$ and $n(A \cup B) = 40$ then $n(A \cap B)$ is?
A. 20 C. 30
B. 40 D. 10
- The relation between sets A, B, C as shown by Venn diagram is _____.
A. A is subset of B and B is subset of C
B. C is not a subset of A and A is subset of B
C. C is subset of B and B is subset of A
D. None of the mentioned
- Let:
 A : All badminton player are good sportsperson.
 B : All person who plays cricket are good sportsperson.
Let X denotes set of all badminton players, Y of all cricket players, Z of all good sportsperson. Then which of the following statements is correct?
A. Z contains both X and Y
B. Z contains X and Y is outside
C. X contains Y and Z
D. None of the mentioned
- If $n(A)=10, n(B)=30, n(C)=50$ and if set A, B, C are pairwise disjoint then which of the following is correct?
A. $n(A \cup B)=0$
B. $n(B \cup C)=0$
C. $n(A \cup B \cup C)=90$
D. All of the mentioned
- If $f(x) = -2x + 8$, find $f^{-1}(1)$.
A. 4.5 C. 3.5
B. 0.22... D. 0



22. If $f(x) = 1/(2 - \sin 3x)$, find $f^{-1}(x)$.
- A. $(1/3)\sin^{-1}[x/(2x - 1)]$
B. $(1/3)\sin[(2x - 1)/x]$
C. $(1/3)\sin^{-1}[(2x - 1)/x]$
D. $2 - \sin 3x$
23. Four friends went shopping for school clothes. Kim bought 5 shirts, Jill bought 4 shirts, Leslie bought 6 shirts, and Crystal bought only 1 shirt. Which answer choice represents the mean of the shirts purchased?
- A. 2
B. 3
C. 4
D. 6
24. What is the median of the following numbers? 1, 2, 2, 8, 9, 14
- A. 6
B. 5
C. 2
D. 13
25. Which set of data has a mean of 15, a range of 22, a median of 14, and a mode of 14?
- A. 25, 15, 14, 3, 7
B. 3, 14, 19, 25, 14
C. 14, 22, 14, 15, 4
D. 14, 22, 15, 15, 9
26. Which number is not the mean, median, or mode of the data set 4, 3, 15, 11, 3, 8, 7, 5?
- A. 6
B. 3
C. 7
D. 5
27. The following are scores made on a math test. 80, 90, 90, 90, 85, 60, 70, 75, 85, 90, 60, 80 What is the mode of these scores?
- A. 70
B. 90
C. 80
D. 60
28. The following are scores made on a math test. 80, 90, 90, 85, 60, 70, 75, 85, 90, 60, 80 What is the median of these scores?
- A. 85
B. 70
C. 80
D. 90
29. The following are scores made on a math test. 80, 90, 90, 85, 60, 70, 75, 85, 90, 60, 80 What is the range of these scores?
- A. 90
B. 30
C. 50
D. 70
30. The mode score on the 6th grade math test was 94! Which of these interpretations must be correct?
- A. More students received a 94 than any other score.
B. A score of 91 was slightly below average.
C. 99 is the highest score in the class.
D. No one scored below a 50.
31. What is the mode of the following numbers? 1, 2, 2, 8, 9, 14
- A. 14
B. 8
C. 9
D. 2
32. What is the mean of the following numbers? 1, 2, 2, 8, 9, 14
- A. 13
B. 5
C. 2
D. 6
33. Find the median of the set of numbers: 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10.
- A. 55
B. 1
C. 10
D. 5.5
34. Find the median of the set of numbers: 21, 3, 7, 17, 19, 31, 46, 20 and 43.
- A. 19
B. 3
C. 20
D. 167
35. Find the median of the set of numbers: 100, 200, 450, 29, 1029, 300 and 2001.
- A. 300
B. 7
C. 29
D. 4,080
36. The following represents age distribution of students in an elementary class. Find the mode of the values: 7, 9, 10, 13, 11, 7, 9, 19, 12, 11, 9, 7, 9, 10, 11.
- A. 7
B. 10
C. 9
D. 11
37. Find the mode from these test results: 90, 80, 77, 86, 90, 91, 77, 66, 69, 65, 43, 65, 75, 43, 90.
- A. 43
B. 65
C. 77
D. 90
38. Find the mode from these test results: 17, 19, 18, 17, 18, 19, 11, 17, 16, 19, 15, 15, 15, 17, 13, 11.
- A. 15
B. 17
C. 11
D. 19
39. Find the mean of these set of numbers: 100, 1050, 320, 600 and 150.
- A. 333
B. 440
C. 444
D. 320
40. The following numbers represent the ages of people on a bus: 3, 6, 27, 13, 6, 8, 12, 20, 5, 10. Calculate their mean of their ages.
- A. 11
B. 9
C. 6
D. 110
41. These numbers are taken from the number of people that attended a particular church every Friday for 7 weeks: 62, 18, 39, 13, 16, 37, 25. Find the mean.
- A. 25
B. 62
C. 210
D. 30
42. If $f(x) = x \sec x$, then $f(0) =$
- A. -1
B. 1
C. 0
D. $\sqrt{2}$
43. If $f(x) = \tan^{-1}(x)$ and $g(x) = \tan(x)$, find $(g \circ f)(x)$.
- A. $\tan^{-1} x \tan x$
B. x
C. $\tan^{-1} x \cot x$
D. $\tan^{-1} x \sin x$
44. If $f(x) = x - 2$ and $g(x) = \sqrt{x^2 + 1}$, then $(g \circ f)(x)$ is equal to
- A. $\sqrt{x^2 + 1} - 2$
B. $\sqrt{x^2 - 4x + 5}$



- C. $x^2 - 1$
D. $x^2 - 4x + 5$

45. If $f(x) = \sqrt{x + 4}$, then $f(x^2 + 4) =$

- A. $x^2 - 8$ C. $x - 8$
B. $x + 8$ D. $x^2 + 8$

46. If $f(x) = \sqrt{2\cos(2\pi \sin x)}$, then $f(\pi/2)$ is equal to ____.

- A. -1 C. 0
B. 1 D. $\sqrt{2}$

47. According to percentiles, the median to measured must lie in ____.

- A. 80th C. 40th
B. 50th D. 100th

48. The median, mode, deciles and percentiles are all considered as measures of ____.

- A. mathematical averages
B. population averages
C. sample averages
D. averages of position

49. A mutual fund achieved the following rates of growth over an 11-month period: {3% 2% 7% 8% 2% 4% 3% 7.5% 7.2% 2.7% 2.09%}
Determine the 5th decile from the data.

- A. 4% C. 3%
B. 2% D. 7%

50. Given the following distribution of returns, determine the lower quartile: {10% 23% 12% 21% 14% 17% 16% 11% 15% 19%}

- A. 11% C. 12%
B. 11.5% D. 19%

END